

Linear Algebra - CS - 3A & 3B

AH Sheikh

Problem Sheet - Week 2

TOPICS: Solving LS using Cramer Rule, Substitution and (Gaussian) Elimination Method.

NOTE: Attempt problem as asked. No need to do computation if only explanation is asked.

1 Cramer's Rule

1. **(2x2 System)** Solve using Cramer's Rule:

$$\begin{cases} 3x - y = 5 \\ 2x + 4y = 8 \end{cases}$$

2. **(2x2 System)** Solve using Cramer's Rule:

$$\begin{cases} 5x + 2y = 7 \\ -x + 3y = 6 \end{cases}$$

3. **(3x3 System)** Solve using Cramer's Rule:

$$\begin{cases} x + y - z = 2 \\ 2x - y + z = 3 \\ x + 2y + z = 7 \end{cases}$$

4. **(3x3 System)** Solve using Cramer's Rule:

$$\begin{cases} x + 2y + 3z = 14 \\ 2x + y + 2z = 10 \\ 3x + 2y + z = 14 \end{cases}$$

5. **(Conceptual)** Under what condition does Cramer's Rule fail to produce a unique solution? Explain why, and give an example of a system where this occurs.

2 Substitution Method

1. **(2x2 System)** Solve using substitution:

$$\begin{cases} x = 3y - 4 \\ 2x + y = 13 \end{cases}$$

2. **(2x2 System)** Solve using substitution:

$$\begin{cases} 4x - 3y = 2 \\ 2x + y = 6 \end{cases}$$

3. **(3x3 System)** Solve using substitution:

$$\begin{cases} x + y + z = 6 \\ y - z = 1 \\ 2x + y = 7 \end{cases}$$

4. **(3x3 System)** Solve using substitution:

$$\begin{cases} x + y = 5 \\ y + z = 7 \\ z + x = 8 \end{cases}$$

5. **(Conceptual)** When is the substitution method more efficient than elimination or Cramer's Rule? Describe at least two scenarios and justify your answer.

3 Elimination Method

1. **(2x2 System)** Solve using elimination:

$$\begin{cases} 2x + 3y = 8 \\ 3x - y = 1 \end{cases}$$

2. **(2x2 System)** Solve using elimination:

$$\begin{cases} 5x - 2y = 4 \\ 2x + 3y = 13 \end{cases}$$

3. **(3x3 System)** Solve using elimination:

$$\begin{cases} x + 2y - z = 1 \\ 2x - y + z = 6 \\ x + y + z = 6 \end{cases}$$

4. **(3x3 System)** Solve using elimination:

$$\begin{cases} 2x - y + 3z = 9 \\ x + 4y - z = -2 \\ 3x + y - 2z = 5 \end{cases}$$

5. **(Conceptual)** Explain the relationship between the elimination method and Gaussian elimination. How are they similar, and what advanced concept does elimination lead to in linear algebra?